DIGITAL FORENSIC SCENERE CONSTRUCTION AND EVIDENCE MANAGEMENT USING IMAGE BASED 3D RECONSTRUCTION AND COMPUTER VISION

A PROJECT REPORT

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in partial fulfillment for the requirements for the award degree of

Bachelor of Engineering

20CS8504 - PROJECT WORK

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

K RAMAKRISHNAN COLLEGE OF TECHNOLOGY (AUTONOMOUS)

SAMAYAPURAM – 621112

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BONAFIDE CERTIFICATE

The work embodied in the present project report entitled "**DIGITAL FORENSIC SCENE RECONSTRUCTION AND EVIDENCE MANAGEMENT USING IMAGE BASED 3D RECONSTRUCTION AND COMPUTER VISION**" has been carried out by the students ,**SARASWATHY P L, SHANMUGA SHREE M, SHARLY PRICILLA A, SRIMATHI K**, The work reported herein is original and we declare that the project is their own work, except where specifically acknowledged, and has not been copied from other sources or been previously submitted for assessment.

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INTERNAL EXAMINER **EXTERNAL EXAMINER**

ABSTRACT

Digital forensic investigation plays a crucial role in modern criminal analysis by enabling investigators to examine and interpret digital evidence collected from crime scenes. Traditional crime scene documentation methods rely on photographs, sketches, and written reports, which may lack accuracy and spatial understanding. To overcome these limitations, Digital Forensic Scene Reconstruction uses advanced technologies to recreate crime scenes in a virtual environment. The proposed system focuses on reconstructing crime scenes digitally using images and computational techniques. The system improves the accuracy, reliability, and efficiency of forensic investigations by providing an interactive platform for analyzing crime scenes. Investigators can revisit the reconstructed environment, examine evidence placement, and better understand the sequence of events that occurred during the crime. Overall, the Digital Forensic Scene Reconstruction system enhances the investigation process by integrating modern digital technologies with forensic science.

Keywords: Digital Forensics, 3D Crime Scene Reconstruction, Computer Vision, Structure from Motion (Sfm), Evidence Management

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SIGNATURE

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LIST OF ABBREVIATIONS

AI	-	Artificial Intelligence
AR	-	Augmented Reality
VR	-	Virtual Reality
API	-	Application Programming Interface
Gui	-	Graphical User Interface
DB	-	Database
SQL	-	Structured Query Language
CPU	-	Central Processing Unit
ML	-	Machine Learning
CV	-	Computer Vision
SFM	-	Structure from Motion
SIFT	-	Scale Invariant Feature Transform
ORB	-	Oriented FAST and Rotated BRIEF
HTTP	-	Hypertext Transfer Protocol
JSON	-	JavaScript Object Notation
REST	-	Representational State Transfer
PNG	-	Portable Network Graphics
JPEG	-	Joint Photographic Experts Group
IDE	-	Integrated Development Environment
GPU	-	Graphics Processing Unit

CHAPTER 1

INTRODUCTION

1.1 DESCRIPTION

Crime scene reconstruction is a scientific method used to determine the sequence of events that occurred during a crime. Traditional methods of documenting crime scenes involve photographs, sketches, and manual documentation. While these methods provide useful information, they often fail to capture the complete spatial structure of the scene.

Digital forensic scene reconstruction is a modern approach that uses computer technologies to recreate crime scenes in a virtual environment. By using digital images and computational algorithms, investigators can generate 3D representations of crime scenes. The proposed Digital Forensic Scene Reconstruction System aims to assist investigators in recreating crime scenes digitally using image data. The system processes uploaded crime scene images and generates a reconstructed digital model.

In recent years, the rapid development of digital technologies such as computer vision, artificial intelligence, and image processing has significantly improved the accuracy and efficiency of forensic investigations. Digital reconstruction enables investigators to analyze complex crime scenes that may otherwise be difficult to interpret using traditional documentation methods. By digitally preserving the crime scene, investigators can revisit and examine the environment multiple times without disturbing the original evidence.

Furthermore, digital reconstruction systems provide an interactive platform where investigators can examine different perspectives of the crime scene. This helps in identifying hidden evidence and understanding how events might have occurred. As a result, digital forensic reconstruction has become an important tool in modern forensic science.

1.2 PROBLEM STATEMENT

In modern forensic investigations, accurate documentation and analysis of crime scenes are critical for identifying evidence and reconstructing events. Traditional methods such as photographs, sketches, and written reports are widely used; however, they are limited in representing the complete spatial structure of the crime scene. These methods often lack depth information and may lead to incomplete understanding or misinterpretation of evidence.

Manual reconstruction of crime scenes is time-consuming and depends heavily on the investigator's expertise, which can result in inconsistencies and reduced accuracy. Additionally, existing systems that use advanced technologies like laser scanning are expensive and not easily accessible to all forensic departments.

There is a need for an efficient, cost-effective, and automated system that can accurately reconstruct crime scenes in a three-dimensional format using readily available image data. Such a system should also provide interactive visualization and secure management of forensic evidence to improve investigation accuracy and support decision making in legal proceedings.

Furthermore, the lack of interactive visualization in existing systems makes it difficult for investigators, legal authorities, and juries to clearly understand complex crime scenes. The absence of centralized and secure evidence management systems also creates challenges in storing, retrieving, and sharing forensic data efficiently. As investigations become more complex, there is a growing need for advanced digital solutions that can provide accurate, real-time visualization and reliable data handling. Therefore, developing a system that integrates 3D reconstruction with effective evidence management is essential to enhance the overall efficiency and reliability of forensic investigations.

1.2 DIGITAL FORENSICS DOMAIN

Digital forensics refers to the process of collecting, analyzing, and preserving digital evidence for investigation and legal purposes. It is widely used in cybercrime investigations and digital evidence analysis.

The field of digital forensics includes several subdomains such as computer forensics, network forensics, mobile device forensics, and multimedia forensics. Each of these areas focuses on analyzing different types of digital evidence to support criminal investigations.

1.2.1 Digital Evidence

Digital evidence includes data stored or transmitted electronically that can be used during an investigation. Examples include digital photographs, surveillance videos, emails, system logs, and files stored on electronic devices.

Digital evidence plays a critical role in modern investigations because many crimes involve digital devices either directly or indirectly. For example, while mobile phones may contain messages or location data related to the crime.

Digital evidence must be handled carefully to maintain its integrity and ensure that it can be used in legal proceedings. Investigators follow strict procedures for collecting, preserving, and analyzing digital data to avoid altering or damaging the evidence.

1.2.2 Crime Scene Documentation

Crime scene documentation involves recording the condition of a crime scene and collecting relevant evidence. Accurate documentation is essential because it preserves the details of the scene before any evidence is removed or disturbed.

Common methods used for crime scene documentation include:

- Photography
- Video recording
- Sketching
- Written reports

These methods provide valuable information but may not fully represent the spatial relationships within the scene. For instance, photographs capture only two-dimensional views, and sketches may depend on the interpretation of the investigator. As a result, some important details may be missed or difficult to interpret.

Digital reconstruction techniques help overcome these limitations by creating a comprehensive digital representation of the crime scene. This allows investigators to examine the scene from different angles and perspectives.

1.2.3 3D Crime Scene Reconstruction

3D reconstruction techniques allow investigators to recreate crime scenes digitally. These techniques use multiple images taken from different angles to generate a three-dimensional model of the environment.

Technologies used for 3D reconstruction include photogrammetry, computer vision, and laser scanning. Photogrammetry uses overlapping images to calculate depth and create accurate 3D models, while laser scanners capture precise measurements of the environment.

3D reconstruction provides investigators with a realistic representation of the crime scene that can be explored interactively. Additionally, reconstructed scenes can be used as visual evidence in courtrooms. Judges and juries can better understand complex cases when presented with clear visual representations of the crime scene.

1.3 AIM AND OBJECTIVE

1.3.1 Aim

The aim of this project is to develop a digital forensic scene reconstruction system that uses computer vision techniques to generate accurate 3D models of crime scenes from images, provides interactive visualization for better analysis, and ensures secure management of forensic evidence to support efficient investigation and decision-making.

1.3.2 Objectives

- To design and develop a user-friendly application that allows users to capture, upload, and manage crime scene images.
- To implement an image acquisition module that captures high quality images from cameras or mobile devices for reconstruction.
- To perform image preprocessing techniques such as filtering, noise reduction, and normalization to enhance image quality.
- To apply feature extraction algorithms such as SIFT and ORB to identify key points in images for matching.
- To develop a 3D reconstruction module using Structure-from-Motion (SfM) techniques to generate 3D models from multiple images.

1.4 SCOPE OF THE PROJECT

The scope of this project focuses on developing a digital forensic system that reconstructs crime scenes into a three-dimensional format using images captured from cameras or mobile devices. The system enables users to capture or upload multiple images of a crime scene, which are then processed using computer vision techniques to generate an accurate 3D model.

The project utilizes image preprocessing, feature extraction, and Structure-from-Motion algorithms to reconstruct the spatial structure of the scene. The generated 3D model allows investigators to visualize the crime scene interactively.

In addition, the system includes an evidence management module that securely stores images, reconstructed models, and related data in a database. This ensures easy access, retrieval, and organization of forensic information. The proposed system improves the efficiency, accuracy, and visualization of crime scene analysis and can be used in forensic investigations, law enforcement agencies, and legal proceedings.

CHAPTER 2

LITERATURE SURVEY

2.1 DIGITAL PHOTOGRAMMETRY FOR CRIME SCENE RECONSTRUCTION

Smith and Johnson (2016) conducted an extensive study on the use of digital photogrammetry techniques for reconstructing crime scenes. Their research focused on capturing multiple overlapping photographs from different viewpoints and processing them using photogrammetric algorithms to generate accurate three-dimensional models.

The authors explained that photogrammetry allows investigators to recreate the spatial layout of a crime scene without disturbing physical evidence. This technique also helps in preserving the scene digitally so that investigators can revisit and analyze the environment even after the physical scene has been cleared. The study emphasized that photogrammetry provides a cost-effective alternative to traditional reconstruction methods such as manual sketches and laser scanning. By integrating multiple technologies, investigators can develop more comprehensive digital models that provide deeper insights into the sequence of events that occurred during a crime.

The researchers demonstrated that photogrammetric models can provide detailed measurements, object positions, and scene geometry, which are essential for forensic analysis. Furthermore, the reconstructed models can be used in courtroom presentations to clearly demonstrate the relationship between evidence and the surrounding environment. The authors concluded that digital photogrammetry significantly improves crime scene documentation and enhances the accuracy of forensic investigations.

2.2 3D LASER SCANNING FOR FORENSIC INVESTIGATION

Brown and Davis (2017) investigated the application of 3D laser scanning technology in forensic crime scene reconstruction. Their research focused on using laser scanners to capture highly accurate spatial measurements of crime scenes and convert them into digital point cloud models. The authors explained that laser scanning technology works by emitting laser beams that measure the distance between the scanner and surrounding objects.

This allows the system to create highly detailed three-dimensional models of the environment with precise measurements. The researchers highlighted that laser scanning provides investigators with the ability to capture an entire crime scene within minutes, reducing the risk of losing important evidence. Additionally, the digital models created through laser scanning allow investigators to perform detailed analysis, measure distances, and reconstruct the sequence of events that occurred during the crime. However, the study also pointed out certain limitations such as the high cost of laser scanning equipment and the need for specialized training to operate these systems.Despite these challenges, the authors concluded that 3D laser scanning technology has become one of the most powerful tools in modern forensic investigations due to its high level of accuracy and reliability.

However, the study also pointed out certain limitations such as the high cost of laser scanning equipment and the need for specialized training to operate these systems. Environmental factors such as reflective surfaces, poor lighting conditions, and obstacles may also affect the quality of the scanned data. Despite these challenges, the authors concluded that 3D laser scanning technology has become one of the most powerful tools in modern forensic investigations due to its high level of accuracy, reliability, and ability to create detailed digital representations of crime scenes.

2.3 COMPUTER VISION TECHNIQUES IN FORENSIC ANALYSIS

Zhang and Lee (2018) explored the application of computer vision algorithms in forensic crime scene analysis. Their research focused on developing automated techniques that can analyze digital images and extract important information from crime scenes. Computer vision methods such as feature detection, object recognition, and image matching were used to identify objects and reconstruct spatial relationships within the scene. The researchers explained that computer vision algorithms can detect key points in images and match them across multiple photographs to estimate the position and orientation of objects. This information can then be used to reconstruct a three dimensional model of the crime scene.

The study demonstrated that computer vision techniques significantly reduce the time required for manual analysis and improve the accuracy of reconstruction results. In addition, the automated detection of objects and evidence helps investigators focus on important details that may otherwise be overlooked. These techniques also enable investigators to identify patterns and relationships between different elements present in the crime scene, which may help in understanding how the crime occurred.The study also suggested that computer vision techniques can be integrated with other technologies such as photogrammetry, artificial intelligence, and machine learning to create more advanced forensic investigation systems.

Furthermore, the researchers highlighted that computer vision techniques can enhance the documentation and presentation of forensic evidence in legal proceedings. By generating detailed visual outputs such as annotated images and reconstructed 3D models, these techniques allow investigators to present complex crime scene information in a clear and understandable manner. This improves communication between forensic experts, investigators, and legal authorities. Additionally, advancements in real-time image processing and automated analysis are expected to further improve the speed and reliability of forensic investigations. As computer vision technology continues to evolve, it has the potential to play a crucial role in developing intelligent forensic systems that provide accurate, efficient, and scalable solutions for crime scene analysis.

2.4 ARTIFICIAL INTELLIGENCE FOR DIGITAL FORENSICS

Kumar and Patel (2019) proposed the use of artificial intelligence techniques in digital forensic investigations. Their research highlighted the growing importance of machine learning algorithms in analyzing large volumes of digital evidence generated from modern devices. The authors explained that AI-based systems can automatically analyze digital data such as images, videos, and logs to identify patterns and extract meaningful information related to criminal activities.

The study showed that AI-based forensic systems can reduce manual effort, improve analysis speed, and minimize human errors during investigations. Traditional forensic analysis often requires investigators to manually examine large amounts of digital data, which can be time-consuming and prone to mistakes. Artificial intelligence algorithms can process this data automatically and identify important information more quickly and accurately.

Another important aspect discussed in the research is the integration of artificial intelligence with image processing and reconstruction technologies. AI algorithms can automatically detect objects within images, classify different types of evidence, and assist in generating digital models of crime scenes. These intelligent systems can also provide recommendations to investigators regarding possible interpretations of the evidence.

The authors concluded that artificial intelligence technologies have the potential to revolutionize forensic investigations by providing intelligent tools that assist investigators in analyzing digital evidence more effectively. By combining AI with computer vision, machine learning, and digital reconstruction techniques, future forensic systems can provide faster, more accurate, and more reliable crime scene analysis.

2.5 3D MODELING TECHNIQUES FOR CRIME SCENE VISUALIZATION

Garcia and Thompson (2020) conducted research on the importance of three-dimensional modeling techniques in forensic crime scene visualization. Their work focused on developing visualization systems that allow investigators to interact with reconstructed digital models of crime scenes. The researchers explained that 3D visualization tools provide an immersive environment where investigators can explore the scene from different perspectives and analyze the spatial relationships between objects and evidence. These visualization systems enable users to rotate, zoom, and navigate through the reconstructed environment, making it easier to understand complex crime scenarios.

The study demonstrated that interactive visualization plays a crucial role in forensic investigations by allowing investigators to examine the scene in greater detail. Traditional documentation methods such as photographs and sketches often provide limited views of the crime scene, which can make it difficult to interpret spatial relationships accurately. In contrast, 3D modeling techniques create a complete representation of the environment, allowing investigators to analyze evidence placement and object positioning more effectively.

Additionally, the study highlighted that modern 3D modeling software can integrate data from multiple sources such as photographs, laser scans, and digital measurements. This integration improves the accuracy and realism of the reconstructed models. The researchers concluded that 3D visualization systems not only enhance the investigation process but also improve communication between investigators, forensic experts, and legal authorities. Furthermore, the researchers emphasized that 3D modeling techniques can significantly enhance the presentation of forensic evidence in courtrooms. By providing realistic and interactive visual representations, investigators can clearly demonstrate the sequence of events and the spatial arrangement of objects within the crime scene. The study also suggested that future advancements in 3D visualization, such as integration with virtual reality technologies, can further improve the immersive experience and accuracy of crime scene analysis, making forensic investigations more effective and reliable.

2.6 STRUCTURE FROM MOTION FOR SCENE RECONSTRUCTION

Ahmed and Hassan (2021) introduced the Structure from Motion (SfM) technique for reconstructing three-dimensional environments using digital images. Their research focused on analyzing multiple photographs taken from different viewpoints and estimating camera positions to generate accurate 3D models. The authors explained that SfM algorithms detect common feature points in images and use mathematical calculations to determine the spatial structure of the scene.

The study demonstrated that Structure from Motion works by identifying matching points across multiple images and calculating the relative position of cameras that captured those images. By analyzing these relationships, the algorithm reconstructs the three-dimensional geometry of the environment. This approach allows investigators to reconstruct crime scenes without the need for expensive hardware such as laser scanners.

The authors concluded that Structure from Motion has become one of the most widely used techniques in digital forensic reconstruction due to its flexibility, accuracy, and affordability. They also suggested that combining SfM with computer vision and artificial intelligence technologies could further improve the efficiency and reliability of reconstruction systems. Furthermore, the researchers highlighted that Structure from Motion techniques can significantly improve the efficiency of forensic investigations by enabling rapid reconstruction of crime scenes using only image data. This reduces the dependency on specialized hardware and allows investigators to work with readily available tools such as digital cameras or smartphones. The ability to revisit and analyze reconstructed scenes multiple times also enhances the reliability of the investigation process. The study also suggested that future improvements in SfM algorithms, such as real-time processing and integration with deep learning models, could further enhance reconstruction accuracy and make the technology more suitable for large-scale and complex forensic applications.

2.7 VIRTUAL REALITY FOR CRIME SCENE INVESTIGATION TRAINING

Wilson and Carter (2022) explored the use of virtual reality technology for forensic training and crime scene analysis. Their research focused on developing immersive virtual environments that replicate real crime scenes in a digital format. The authors explained that virtual reality allows investigators to interact with reconstructed scenes in a realistic manner, providing a better understanding of the spatial layout and evidence placement.

The study demonstrated that VR-based training systems help forensic students and investigators develop practical investigation skills without requiring access to real crime scenes. In a virtual environment, trainees can explore the reconstructed scene, collect digital evidence, and practice analyzing different types of crime scenarios. This hands-on experience improves their ability to identify important evidence and understand investigation procedures.Another important advantage of virtual reality systems is the ability to simulate different conditions and scenarios. For example, investigators can test various hypotheses about how a crime occurred by adjusting variables within the virtual environment. This allows them to analyze possible sequences of events and evaluate different interpretations of the evidence.

The study concluded that integrating virtual reality technology with digital reconstruction systems can significantly enhance forensic education and improve the effectiveness of crime scene investigations. Furthermore, the researchers highlighted that virtual reality technology can improve collaboration among investigators and forensic experts by allowing multiple users to access and interact with the same virtual crime scene simultaneously. This collaborative environment enables experts from different locations to analyze evidence, discuss findings, and make informed decisions in real time. The study also suggested that advancements in VR hardware and software, such as improved graphics rendering and motion tracking, can further enhance the realism and effectiveness of virtual training systems. As a result, virtual reality is expected to play a significant role in both forensic training and real-world crime scene analysis in the future.

2.8 AUTOMATED SCENE RECONSTRUCTION USING DEEP LEARNING

Chen and Li (2023) proposed a deep learning approach for automating the crime scene reconstruction process. Their research focused on using convolutional neural networks (CNNs) to analyze crime scene images and detect important objects and structures. The authors developed a deep learning model capable of identifying key features in images and generating a three-dimensional representation of the scene automatically.

The study showed that deep learning algorithms significantly improve the speed and accuracy of reconstruction by reducing the need for manual intervention. Traditional reconstruction methods often require investigators to manually identify feature points and align images, which can be time-consuming and error-prone. Deep learning models can automate these processes by learning patterns from large datasets of images.

The researchers also explained that deep learning systems can perform object detection and classification within crime scene images. For example, the system can automatically identify weapons, furniture, vehicles, and other objects present in the scene. This information can then be used to generate more accurate digital models and assist investigators in analyzing evidence.

As the system is trained with additional data, it becomes better at recognizing patterns and detecting relevant features. This continuous learning capability makes deep learning-based reconstruction systems highly adaptable to different types of crime scenes.

The study concluded that automated reconstruction systems based on deep learning have great potential for improving forensic investigations. By integrating deep learning with computer vision, image processing, and 3D modeling techniques, future forensic systems can provide faster, more accurate, and more intelligent crime scene analysis.

2.9 CLOUD BASED DIGITAL FORENSIC PLATFORMS

Martin and Clarke (2024) introduced a cloud-based digital forensic platform designed to improve collaboration among investigators. Their research focused on developing a system that allows investigators to store, process, and analyze digital evidence using cloud computing technologies. The authors explained that cloud-based systems enable investigators from different locations to access and analyze reconstructed crime scenes simultaneously.

The study highlighted that traditional forensic systems often rely on local storage and computing resources, which can limit accessibility and scalability. Cloud-based platforms address these challenges by providing centralized storage and powerful computing capabilities that can handle large volumes of digital data. Investigators can upload crime scene images, reconstruction models, and other digital evidence to the cloud, where it can be securely stored and analyzed.

Another important advantage of cloud-based systems is improved collaboration. Multiple investigators and forensic experts can access the same digital models and datasets in real time, allowing them to work together more effectively. This collaborative approach is particularly useful in complex investigations that involve experts from different disciplines.

The researchers also emphasized the importance of security and data integrity in cloud-based forensic platforms. Advanced encryption techniques and access control mechanisms are used to ensure that sensitive evidence is protected from unauthorized access.

The study concluded that cloud-based forensic platforms represent the future of digital forensic investigations by providing flexible, scalable, and collaborative environments for analyzing crime scenes. As cloud technology continues to develop, it is expected to play a significant role in improving the efficiency and accessibility of forensic investigation systems.

2.10 IMAGE BASED 3D CRIME SCENE RECONSTRUCTION USING COMPUTER VISION

Sharma and Rao (2023) proposed an image-based 3D crime scene reconstruction system using computer vision techniques to improve forensic investigation. Their research focused on generating accurate 3D models of crime scenes using multiple images captured from different angles. The authors explained that image-based reconstruction methods, such as photogrammetry and Structure from Motion (SfM), enable the automatic creation of 3D models without requiring expensive hardware.

The study highlighted that traditional crime scene documentation methods, including photographs and sketches, often lack spatial depth and may lead to incomplete understanding of the scene. Image-based reconstruction addresses these limitations by providing a detailed 3D representation that allows investigators to visualize the spatial arrangement of evidence more effectively.

Another important advantage of the proposed approach is cost-effectiveness. Since the system relies on standard cameras or mobile devices for capturing images, it eliminates the need for specialized equipment such as laser scanners. This makes the technology accessible to a wider range of forensic departments.

The researchers also emphasized the importance of accurate feature detection and matching techniques in improving reconstruction quality. Algorithms such as SIFT and ORB are used to identify key points in images, which are then used to build the 3D structure. Proper image preprocessing and alignment further enhance the accuracy of the final model.

CHAPTER 3

EXISTING SYSTEM

3.1 EXISTING SYSTEM

In recent years, crime scene investigation has relied heavily on traditional documentation techniques such as photography, sketching, and written reports. These methods are commonly used by forensic experts to record the condition of a crime scene and preserve evidence for further analysis. Although these techniques are widely accepted, they often lack the ability to capture the complete spatial structure of the scene. As a result, investigators may face difficulties in understanding the relationships between objects and evidence present at the crime location.

To assist forensic investigations, some digital systems have been developed that utilize advanced technologies such as laser scanning and photogrammetry for crime scene reconstruction. These systems aim to create digital models of crime scenes; however, they often require expensive equipment and specialized expertise. The existing system considered in this work is based on traditional image-based documentation combined with limited digital processing for analysis.

In this system, crime scene images are captured manually using cameras, and basic measurements are recorded to represent the scene. These images are then analyzed by investigators to identify important evidence and reconstruct the sequence of events. However, the reconstruction process is largely manual and depends on the experience and interpretation of the investigator.

Although these methods provide useful information, they lack interactive visualization and do not allow investigators to explore the scene in a dynamic manner. The absence of automated reconstruction and real-time analysis limits the efficiency and accuracy of the investigation process.

Figure 3.1 Existing System

3.2 DISADVANTAGES

- Lack of spatial understanding due to 2D images and sketches.
- Manual documentation is time-consuming and prone to human error.
- Difficult to accurately reconstruct complex crime scenes.
- Limited visualization makes it hard to analyze evidence clearly.
- High dependency on investigator's experience and interpretation.
- Incomplete or inconsistent data collection may affect investigation results.
- No interactive or 360-degree view of the crime scene.
- Difficulty in managing and organizing large amounts of forensic data.
- Risk of data loss or damage due to lack of secure storage systems.
- Limited collaboration between investigators and forensic experts.

CHAPTER 4

PROBLEMS IDENTIFIED

Traditional crime scene reconstruction methods require investigators to manually collect, organize, and analyze large amounts of data. This includes taking photographs from multiple angles, creating sketches, measuring distances, and writing reports. The entire process is time-consuming and may slow down investigations.

Although advanced technologies such as laser scanners and professional 3D modeling tools can produce accurate reconstructions, they are expensive and not accessible to all law enforcement agencies. Many departments, especially in developing regions, cannot afford these specialized tools. In addition, these systems often require trained professionals to operate them, which further increases operational costs.

Another significant challenge is presenting crime scene evidence effectively in court. Judges, lawyers, and juries may find it difficult to interpret crime scene photographs and technical reports. Without a clear visual representation, explaining the sequence of events or the location of evidence can be confusing.

In many cases, crime scene data is stored in separate formats such as image files, paper documents, and digital reports. This fragmented storage makes it difficult for investigators, forensic analysts, and legal authorities to collaborate efficiently. Accessing and sharing information among multiple teams can become complicated and time-consuming.

Manual documentation and analysis of crime scenes increase the risk of human errors. Investigators may accidentally miss important evidence, misinterpret measurements, or record inaccurate information. Even small mistakes in documentation can lead to incorrect conclusions during the investigation process.

CHAPTER 5

PROPOSED SYSTEM

5.1 PROPOSED SYSTEM OVERVIEW

The proposed system aims to develop a digital forensic platform that reconstructs crime scenes into a three-dimensional model using image-based computer vision techniques. In modern forensic investigations, accurate documentation and analysis of crime scenes are essential for understanding events and presenting evidence in legal proceedings. However, traditional methods rely on manual documentation such as photographs and sketches, which may not provide complete spatial understanding. The proposed system introduces an automated approach for reconstructing crime scenes using images captured from cameras or mobile devices. By applying advanced computer vision techniques, the system processes multiple images and generates an accurate 3D representation of the crime scene. This approach improves the efficiency of forensic analysis and enables investigators to visualize the scene interactively.

5.2 IMAGE ACQUISITION

Image acquisition is the first stage of the proposed system, where crime scene images are captured using cameras or mobile devices. The system ensures that images are taken from multiple angles to provide sufficient data for accurate reconstruction.

After acquisition, the images undergo preprocessing to improve their quality. Image preprocessing techniques such as filtering, noise reduction, and normalization are applied to enhance important visual features. These operations help in improving the clarity of the images and prepare them for further processing.

Feature extraction techniques such as SIFT and ORB are then applied to identify key points in the images. These key points are essential for matching features across multiple images and play a crucial role in the 3D reconstruction process.

5.3 3D RECONSTRUCTION USING CV

The proposed system utilizes computer vision techniques to reconstruct the crime scene in three dimensions. The reconstruction process is based on the Structure- from-Motion (SfM) algorithm, which estimates the spatial structure of the scene from multiple 2D images.

In this stage, the system identifies matching features between images and estimates camera positions. Using these correspondences, the system generates a sparse point cloud that represents the basic structure of the scene. Further processing using Multi-View Stereo techniques converts the sparse point cloud into a dense representation.

This 3D reconstruction process enables the system to generate an accurate digital model of the crime scene, allowing investigators to analyze spatial relationships between objects and evidence.

5.4 VISUALIZATION

After reconstruction, the generated 3D model is visualized using an interactive interface. The system allows users to rotate, zoom, and explore the reconstructed crime scene from different angles. This 360-degree visualization helps investigators better understand the spatial arrangement of objects and evidence.

In addition to visualization, the proposed system includes an evidence management module that stores images, 3D models, and related data securely in a database. The system organizes evidence based on case details and ensures that data is protected from unauthorized access.

5.5 PROPOSED SYSTEM ARCHITECTURE

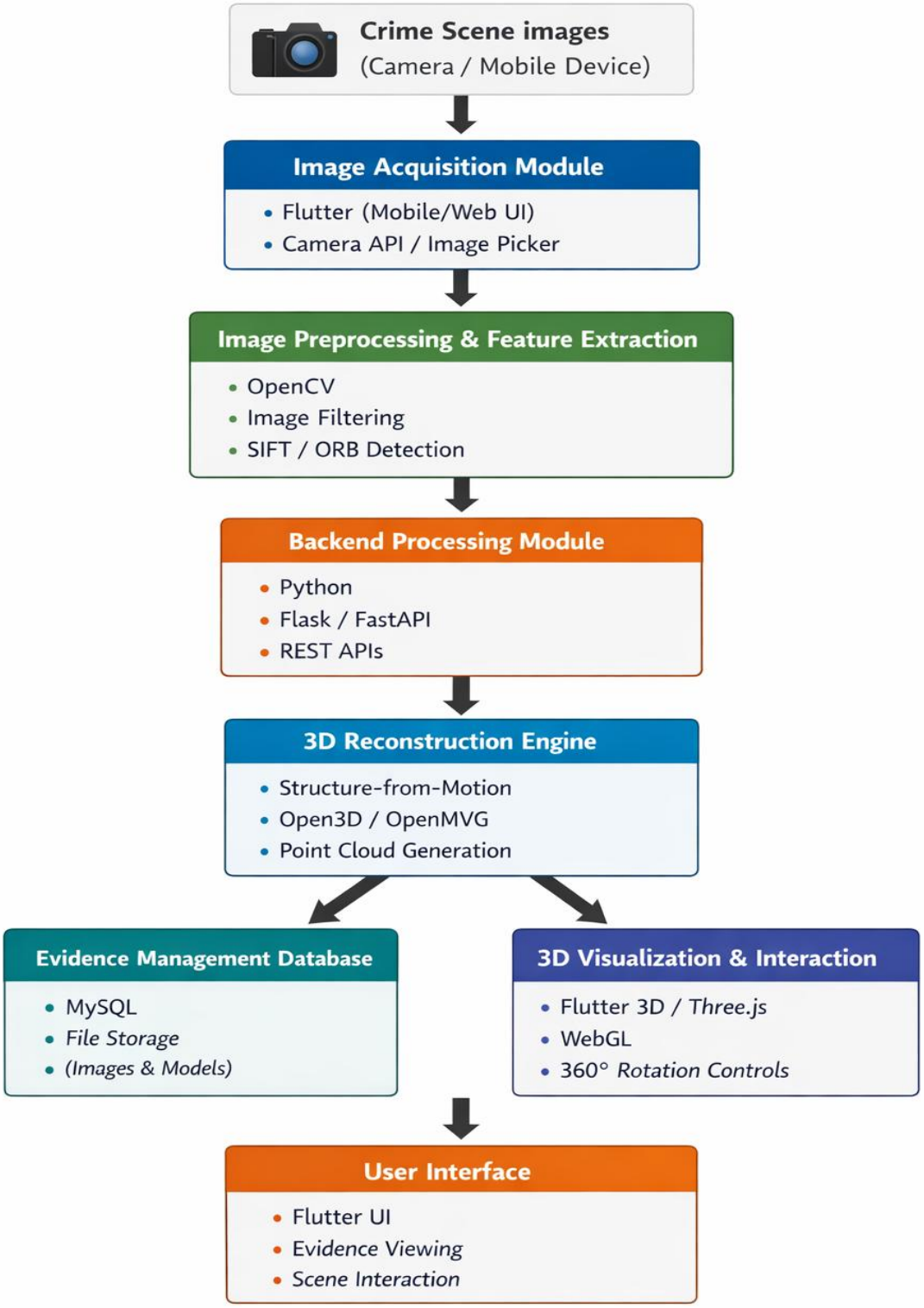

Figure 5.5 Proposed System

The overall workflow of the proposed system begins with capturing crime scene images using a camera or mobile device. The images are then preprocessed to enhance quality and extract important features. These processed images are passed to the reconstruction engine, where 3D models are generated using Structure-from-Motion techniques. The reconstructed model is then visualized in an interactive interface, allowing users to explore the scene. Finally, all images and models are stored securely in the evidence management database for future analysis and reporting.

5.6 ADVANTAGES

- Provides accurate 3D reconstruction of crime scenes using image data.
- Improves spatial understanding with interactive 360-degree visualization.
- Reduces manual effort and time required for crime scene analysis.
- Enhances evidence visualization for better interpretation and decision-making.
- Enables automated feature extraction and reconstruction using computer vision.
- Provides secure storage and efficient management of forensic evidence.
- Reduces dependency on expensive hardware like laser scanners.
- Allows easy sharing and collaboration among investigators and experts.

CHAPTER 6

SYSTEM REQUIREMENTS

6.1 HARDWARE REQUIREMENTS

• Processor

A computer system with a modern processor such as Intel Core i5 or higher is required to run the proposed system efficiently. The processor handles image processing tasks, execution of backend scripts, and 3D reconstruction algorithms. A multi-core processor ensures smooth performance during feature extraction, reconstruction, and visualization processes.

• Memory (RAM)

The system requires a minimum of 8 GB RAM to execute the application without performance issues. Adequate memory is necessary for processing multiple images, running computer vision algorithms, and handling 3D model generation. Sufficient RAM ensures smooth multitasking and faster execution of reconstruction processes.

• Storage

A minimum of 256 GB SSD storage is required to store project files, captured images, reconstructed 3D models, and database records. SSD storage improves system speed and ensures faster access to large image datasets and 3D model files.

• Camera / Input Device

A high-resolution digital camera or smartphone is required to capture crime scene images. The quality of images plays a crucial role in accurate 3D reconstruction, as multiple clear images from different angles are needed for better results.

• Internet Connectivity

A stable internet connection is required for data transfer between the frontend and backend systems. It also supports uploading images, accessing APIs, and enabling communication between different modules of the system.

6.2 SOFTWARE REQUIREMENTS

• Operating System

The proposed system can be developed and executed on Windows or Linux operating systems. These platforms provide the necessary environment to run Python applications, databases, and 3D visualization tools efficiently.

• Frontend (Flutter)

Flutter is used to develop the user interface of the system. It provides a responsive and user-friendly interface for capturing images, viewing evidence, and interacting with the 3D reconstructed scene.

• Backend (Python)

Python is used as the primary programming language for implementing backend processing. It handles image preprocessing, feature extraction.

• Framework (Flask / FastAPI)

Flask or FastAPI is used to develop REST APIs for communication between the frontend and backend. These frameworks manage data transfer, handle user requests, and process image inputs efficiently.

• Database (SQLite / MySQL)

SQLite or MySQL is used for storing forensic data such as images, reconstructed models, and case details. The database ensures secure storage and easy retrieval of information during investigation.

• 3D Processing Libraries

Libraries such as OpenCV, OpenMVG, and Open3D are used for image processing and 3D reconstruction. OpenCV is used for preprocessing and feature extraction, while OpenMVG and Open3D are used for generating and processing 3D

• Development Environment

Visual Studio Code (VS Code) is used as the development environment for implementing the project. It provides tools for coding, debugging, and managing project files efficiently.

CHAPTER 7

SYSTEM IMPLEMENTATIONS

7.1 LIST OF MODULES

The proposed system for digital forensic scene reconstruction and evidence management is implemented using several functional modules. Each module performs a specific task in the overall workflow of the system. The modular design helps in organizing the processing stages efficiently and improves the performance and accuracy of the system. The following modules are used in the proposed system:

- Image Acquisition Module
- Evidence Management Module
- Backend Processing Module
- 3D Reconstruction Module
- 3D Visualization and Interaction Module

7.2 MODULE DESCRIPTION

7.2.1 Image Acquisition Module

The Image Acquisition Module is responsible for capturing and uploading crime scene images into the system. In a forensic investigation, collecting clear and detailed images of the crime scene is a crucial step because these images serve as the primary input for the reconstruction process. Investigators use cameras or mobile devices to capture multiple photographs from different angles and perspectives to ensure that all parts of the crime scene are documented properly.

This module provides a user-friendly interface that allows investigators to upload images easily through a mobile or web application. Technologies such as camera APIs and image picker tools are used to capture images directly from the device or select them from the gallery.

The quality and variety of images captured through this module significantly affect the accuracy of the reconstruction process. Therefore, investigators are encouraged to capture multiple overlapping images of the scene to ensure better feature detection and 3D reconstruction.

7.2.2 Evidence Management Module

The Evidence Management Module is responsible for storing, organizing, and managing all crime scene data within the system. During forensic investigations, large amounts of information such as images, reconstructed models, and investigation records need to be stored securely. This module ensures that all evidence is properly maintained in a structured format.

A database system such as SQLite or MySQL is used to store the collected images, reconstructed 3D models, and associated case information. The module allows investigators to access, update, and retrieve stored data whenever required. Proper data management helps investigators track multiple cases efficiently and prevents loss of important evidence.

Additionally, this module supports the classification and tagging of evidence, making it easier for investigators to search and review specific information related to a case. Secure storage also ensures that the integrity of forensic evidence is maintained throughout the investigation process.

7.2.3 Backend Processing Module

The Backend Processing Module acts as the core processing unit of the system. It handles all the computational tasks required for processing uploaded images and coordinating different modules of the system. This module is developed using backend technologies such as Python with frameworks like Flask or FastAPI.

The backend module uses REST APIs to allow seamless communication between the frontend user interface and the system's internal components. It ensures that the data flow between modules is smooth and efficient while maintaining system security and reliability.

7.2.4 3D Reconstruction Module

The 3D Reconstruction Module is the most important component of the proposed system. Its main function is to generate a three-dimensional model of the crime scene using the uploaded images. The module analyzes multiple images of the scene and identifies common features among them.

Using techniques such as Structure from Motion (SfM), the system estimates camera positions and reconstructs the spatial structure of the environment.

The- reconstructed model accurately represents the layout of the crime scene and the position of objects within it. This digital reconstruction allows investigators to examine the scene in detail even after the physical crime scene has been cleared.

7.2.5 3D Visualization and Interaction Module

The 3D Visualization and Interaction Module allows investigators to view and interact with the reconstructed crime scene in a virtual environment. This module converts the generated 3D model into an interactive visual format that can be displayed on a computer.

Technologies such as 3D viewers or web-based visualization frameworks are used to render the reconstructed scene. Investigators can rotate the model, zoom into specific areas, and explore the scene from different angles. This interactive capability helps investigators better understand the spatial relationships between objects and evidence.

CHAPTER 8
SYSTEM TESTING

8.1 UNIT TESTING

Unit testing is the process of testing individual components or modules of the system separately to verify that each module performs its intended function correctly. In this project, each module such as the Image Acquisition Module, Backend Processing Module, Evidence Management Module, 3D Reconstruction Module, and Visualization Module is tested independently.

The purpose of unit testing is to detect errors at an early stage of development. Developers test functions, input validation, and outputs of each module to ensure they produce the expected results. For example, the image upload functionality is tested to verify that images are successfully captured and stored in the database.

8.2 INTEGRATION TESTING

Integration testing is performed after individual modules have been tested successfully. In this phase, multiple modules are combined and tested together to ensure that they work properly as a unified system.

In the proposed system, integration testing verifies the communication between the frontend interface, backend server, database, and reconstruction engine. For example, after images are uploaded through the interface, the system should correctly transfer them to the backend for processing and then generate the reconstruction output. This testing ensures that data flows smoothly between all modules.

8.3 SYSTEM TESTING

System testing involves testing the entire application as a complete system. The goal of system testing is to verify that the integrated system meets all functional and non-functional requirements.

During system testing, the Digital Forensic Crime Scene Reconstruction System is tested using real-world scenarios. Thistesting ensures that the system behaves as expected and produces accurate outputs under normal operating conditions.

8.4 PERFORMANCE TESTING

Performance testing is conducted to evaluate how well the system performs under different workloads. This includes measuring system response time, processing speed, and overall efficiency. For the proposed system, performance testing ensures that image processing and 3D reconstruction operations are completed within an acceptable time frame. The system is tested with multiple images and larger datasets to verify that it can handle heavy processing tasks without significant delays or system failures.

8.5 USABILITY TESTING

Usability testing evaluates how easy and user-friendly the system is for investigators. Since the system will be used by law enforcement personnel, it is important that the interface is simple, clear, and easy to navigate.

During usability testing, users interact with the system to upload images, view reconstructed models, and access stored evidence. Feedback from users is used to improve the interface design and ensure that the system provides a smooth and efficient user experience.

CHAPTER 9
RESULT AND DISCUSSION

The performance comparison between the existing system and the proposed system clearly demonstrates the effectiveness of the proposed approach in improving crime scene reconstruction and analysis. The evaluation is based on key metrics such as accuracy, processing speed, visualization quality, and data management. The existing system relies on manual documentation and 2D representations, which results in lower accuracy and limited understanding of spatial relationships. In contrast, the proposed system utilizes computer vision techniques and 3D reconstruction methods, leading to a significant improvement in accuracy and reliability of results.

In terms of processing speed, the existing system requires considerable manual effort for data collection and reconstruction, making it time-consuming. The proposed system automates image processing, feature extraction, and reconstruction, thereby reducing the overall time required for analysis. Additionally, visualization quality is greatly enhanced in the proposed system through interactive 3D models, allowing investigators to explore the crime scene from multiple angles. This provides a better understanding of object positions and relationships compared to static 2D images used in the existing system.

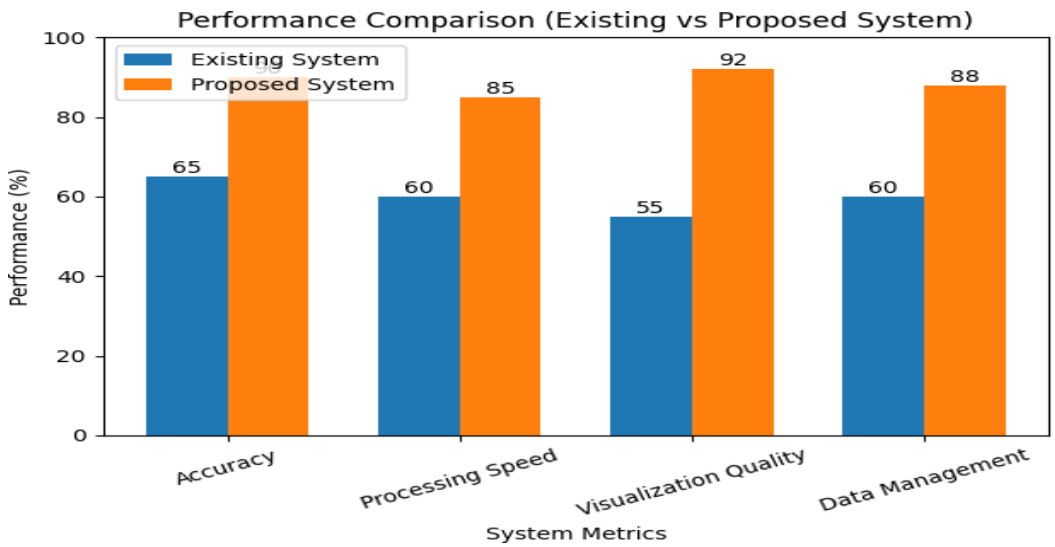

Figure 9.1 Performance Comparison

The accuracy comparison between different methods highlights the effectiveness of the proposed system in crime scene reconstruction. The evaluation is performed by comparing multiple approaches, including basic image processing methods, feature-based techniques, and a basic Structure-from-Motion (SfM) model. The results show that traditional methods provide moderate accuracy due to limited feature extraction and lack of advanced processing capabilities. These methods often struggle to capture complex spatial relationships and may produce less reliable results.

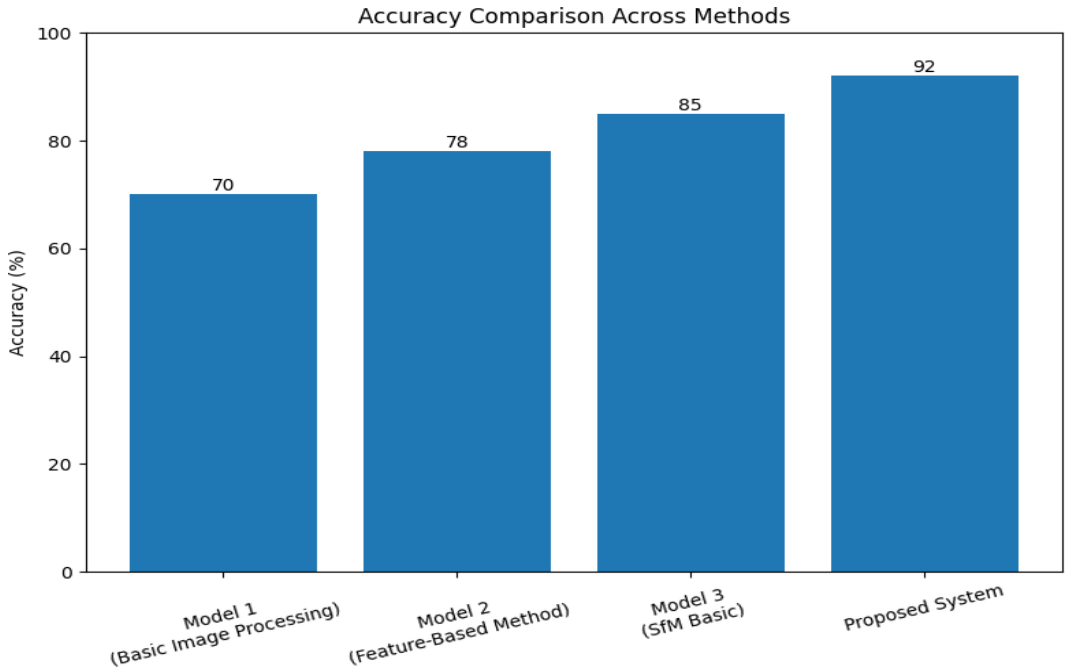

Figure 9.2 Accuracy Comparison

In comparison, the proposed system achieves significantly higher accuracy due to the integration of advanced computer vision techniques such as feature detection, matching, and optimized 3D reconstruction. By using algorithms like SIFT and ORB along with Structure-from-Motion, the system is able to accurately identify key points and reconstruct the scene with better precision. Overall, the accuracy comparison demonstrates that the proposed system outperforms other methods by producing more precise and reliable results. This improvement in accuracy ensures better analysis of crime scenes and supports more effective decision-making during forensic investigations.

CHAPTER 10
CONCLUSION AND FUTURE ENHANCEMENTS

10.1 CONCLUSION

The Digital Forensic Crime Scene Reconstruction System was developed to improve the process of crime scene investigation by utilizing modern digital technologies. Traditional methods of crime scene documentation rely heavily on photographs, sketches, and written reports, which often fail to capture the complete spatial structure of the scene. These limitations can make it difficult for investigators to accurately analyze the sequence of events and the relationship between pieces of evidence.

The proposed system addresses these challenges by providing a digital platform that allows investigators to reconstruct crime scenes using multiple images captured from cameras or mobile devices. The system integrates several modules including image acquisition, evidence management, backend processing, 3D reconstruction, and visualization. These modules work together to process uploaded images and generate a three-dimensional representation of the crime scene.

The testing results show that the system performs its intended functions effectively. The modules operate correctly, the reconstruction process generates useful visual outputs, and the user interface allows investigators to interact with the system easily. Overall, the Digital Forensic Crime Scene Reconstruction System demonstrates how digital technologies can significantly improve crime scene documentation, analysis, and presentation.

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10.2 FUTURE ENHANCEMENTS

Although the proposed system successfully reconstructs crime scenes digitally, several improvements can be made in the future to enhance its functionality and efficiency.

One possible enhancement is the integration of artificial intelligence and machine learning techniques to automatically detect and highlight important evidence within the reconstructed scene. This would assist investigators by reducing manual effort and improving the speed of analysis.

Another future improvement is the incorporation of augmented reality (AR) and virtual reality (VR) technologies. These technologies would allow investigators to immerse themselves in the reconstructed crime scene and analyze evidence in a more realistic and interactive environment.

The system can also be enhanced by supporting real-time crime scene scanning using mobile devices or drones. This would allow investigators to capture images and generate reconstruction models more quickly during field investigations.

Another potential enhancement is the use of cloud-based storage and collaboration tools, which would allow multiple investigators to access and analyze crime scene data from different locations. This would improve teamwork and make it easier to share information between forensic experts and law enforcement agencies.

Additionally, future versions of the system could include advanced measurement and simulation tools that allow investigators to measure distances, simulate possible crime scenarios, and analyze the movement of objects within the reconstructed environment.

By implementing these enhancements, the Digital Forensic Crime Scene Reconstruction System can become a more powerful and efficient tool for modern forensic investigations.

APPENDIX – A
SOURCE CODE

HomePage.tsx

```
import React, { useRef } from 'react';
import { useMember } from '@/integrations';
import { useNavigate } from 'react-router-dom';
import { motion, useScroll, useTransform } from 'framer-motion';
import { Button } from '@/components/ui/button';
import { Shield, Database, Scan, FileText, ArrowRight, Lock, Terminal,
Activity, Cpu, Crosshair, Layers } from 'lucide-react';
import Header from '@/components/Header';
import Footer from '@/components/Footer';
import { Image } from '@/components/ui/image';
export default function HomePage() {
  const { isAuthenticated, actions } = useMember();
  const navigate = useNavigate();
  const features = [
    {
      icon: Shield,
      title: 'Case Management',
              description: 'Create, organize, and track forensic investigations with
        comprehensive case details and officer assignments.'
    },
    {
                <Database className="w-6 h-6 text-primary" />
              </div>
                      <h3 className="font-heading text-2xl font-bold">1. Data
        Ingestion</h3>
            </div>
```

```
<p className="font-paragraph text-foreground/70 mb-6">
  Securely upload hundreds of high-resolution photographs from the scene. The system automatically extracts EXIF data, timestamps, and spatial metadata.
</p>
  <div className="aspect-video relative overflow-hidden border border-primary/20">
 <Image
  src="https://static.wixstatic.com/media/a478a7_1dce1e89d4f84235aaf2d23439e2f67c~mv2.png?originWidth=960&originHeight=512"
  alt="Data ingestion interface"
    className="w-full h-full object-cover opacity-60 mix-blend-luminosity hover:mix-blend-normal transition-all duration-700"
 />
    <div className="absolute inset-0 bg-primary/10 mix-blend-overlay" />
  </div>
 </div>
</motion.div>
<motion.div
 initial={{ opacity: 0, y: 50 }}
 whileInView={{ opacity: 1, y: 0 }}
 viewport={{ once: true, margin: "-20%" }}
 className="relative pl-0 lg:pl-24"
>
  <div className="absolute left-[-5px] top-6 w-3 h-3 rounded-full bg-background border-2 border-secondary hidden lg:block z-10" />
 <div className="bg-code-background/50 border border-secondary/20 p-8 clip-angle backdrop-blur-sm">
  <div className="flex items-center gap-4 mb-6">
```

```
                <div className="w-12 h-12 bg-secondary/10 flex items-center
justify-center border border-secondary/30">
                  <Cpu className="w-6 h-6 text-secondary" />
                </div>
                        <h3 className="font-heading text-2xl font-bold">2.
Photogrammetric Processing</h3>
              </div>
              <p className="font-paragraph text-foreground/70 mb-6">
                Advanced computer vision algorithms analyze overlapping images,
identifying millions of tie points to calculate depth and spatial relationships.
              </p>
                <div className="aspect-video relative overflow-hidden border
border-secondary/20">
                <Image
                  src="https://static.wixstatic.com/media/a478a7_9b15aefe70d04d1
a9a76d21d6768ece4~mv2.png?originWidth=960&originHeight=512"
                  alt="Processing visualization"
                     className="w-full h-full object-cover opacity-60 mix-blend-
luminosity hover:mix-blend-normal transition-all duration-700"
                />
                  <div className="absolute inset-0 bg-secondary/10 mix-blend-
overlay" />
                {/* Simulated processing overlay */}
                <div className="absolute inset-0 tech-grid opacity-30" />
              </div>
            </div>
          </motion.div>

          {/* Step 3 */}
          <motion.div
            initial={{ opacity: 0, y: 50 }}
```

```
            whileInView={{ opacity: 1, y: 0 }}
            viewport={{ once: true, margin: "-20%" }}
            className="relative pl-0 lg:pl-24"
          >
            <div className="absolute left-[-5px] top-6 w-3 h-3 rounded-full bg-background border-2 border-primary hidden lg:block z-10" />

            <div className="absolute inset-0 bg-gradient-to-br from-primary/5 to-transparent opacity-0 group-hover:opacity-100 transition-opacity duration-500" />
            <div className="absolute top-0 left-0 w-full h-[1px] bg-gradient-to-r from-transparent via-primary to-transparent scale-x-0 group-hover:scale-x-100 transition-transform duration-700 origin-left">
            <div className="relative z-10 h-full flex flex-col">
                <div className={`mb-8 flex items-center justify-center bg-background border border-primary/20 ${isLarge ? 'w-20 h-20' : 'w-14 h-14'}`}>
                <feature.icon className={`${isLarge ? 'w-10 h-10' : 'w-6 h-6'} text-primary group-hover:scale-110 transition-transform duration-500`} />
              </div>

              <h3 className={`font-heading font-bold mb-4 ${isLarge ? 'text-4xl' : 'text-2xl'}`}>
                {feature.title}
              </h3>
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                {feature.description}
              </p>
    </div>
  );
}
```

APPENDIX – B

SCREENSHOTS

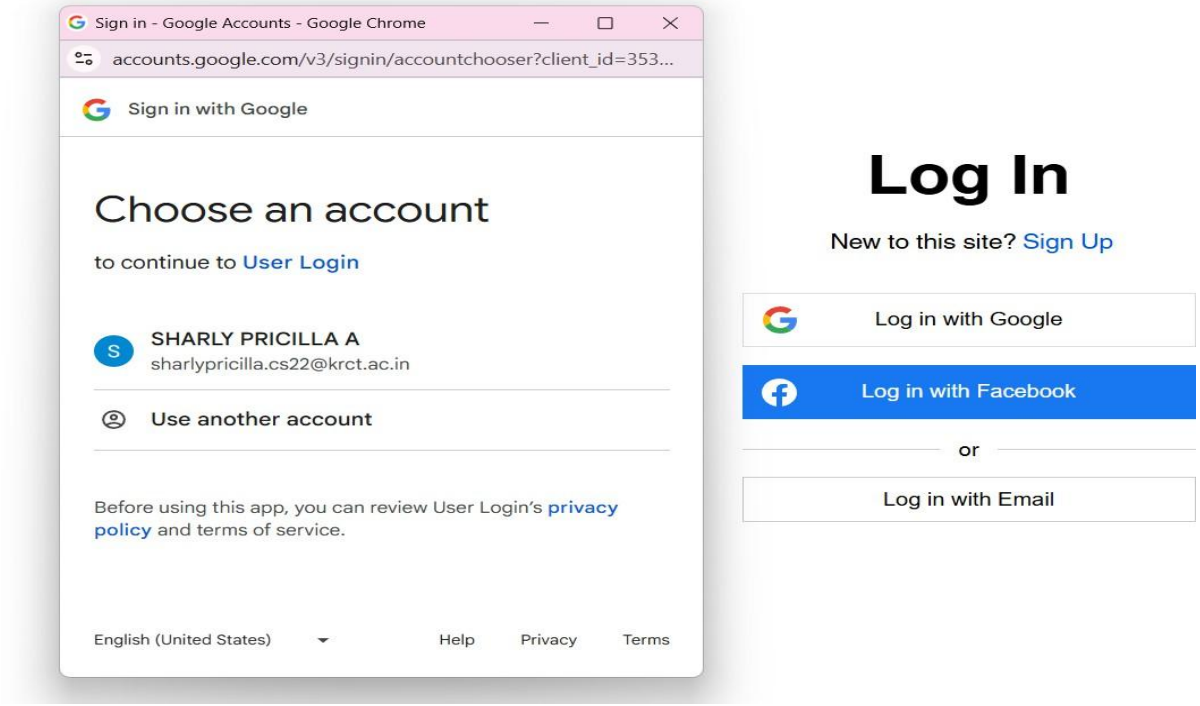

Figure B.1 Login Page

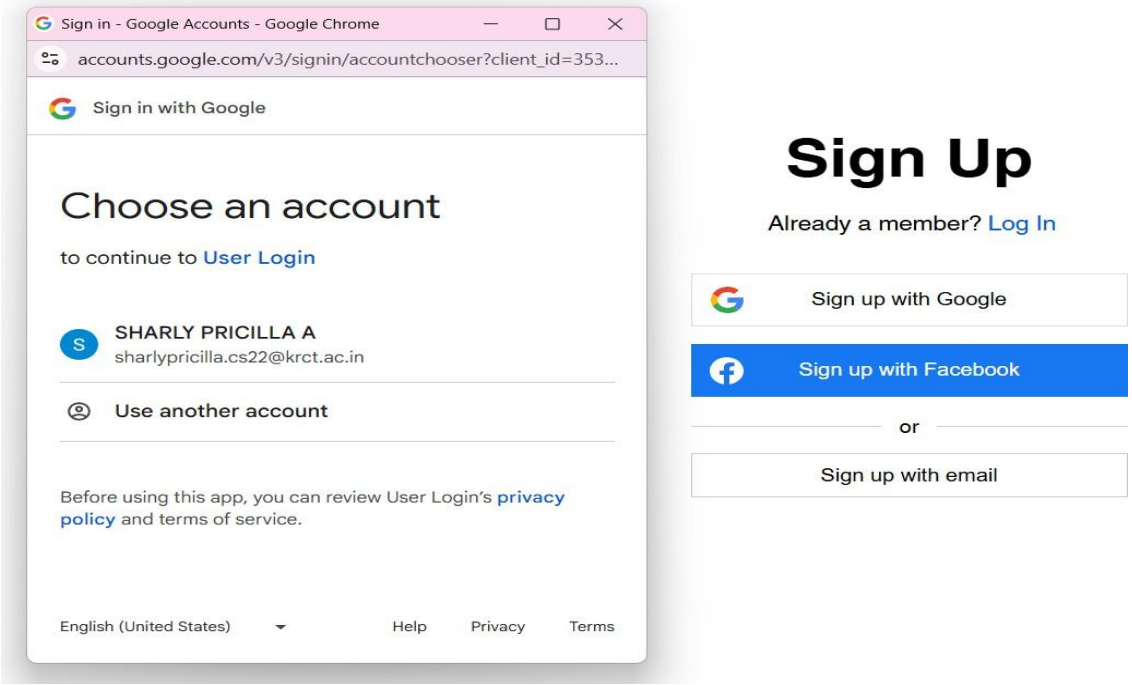

Figure B.2 Signup Page

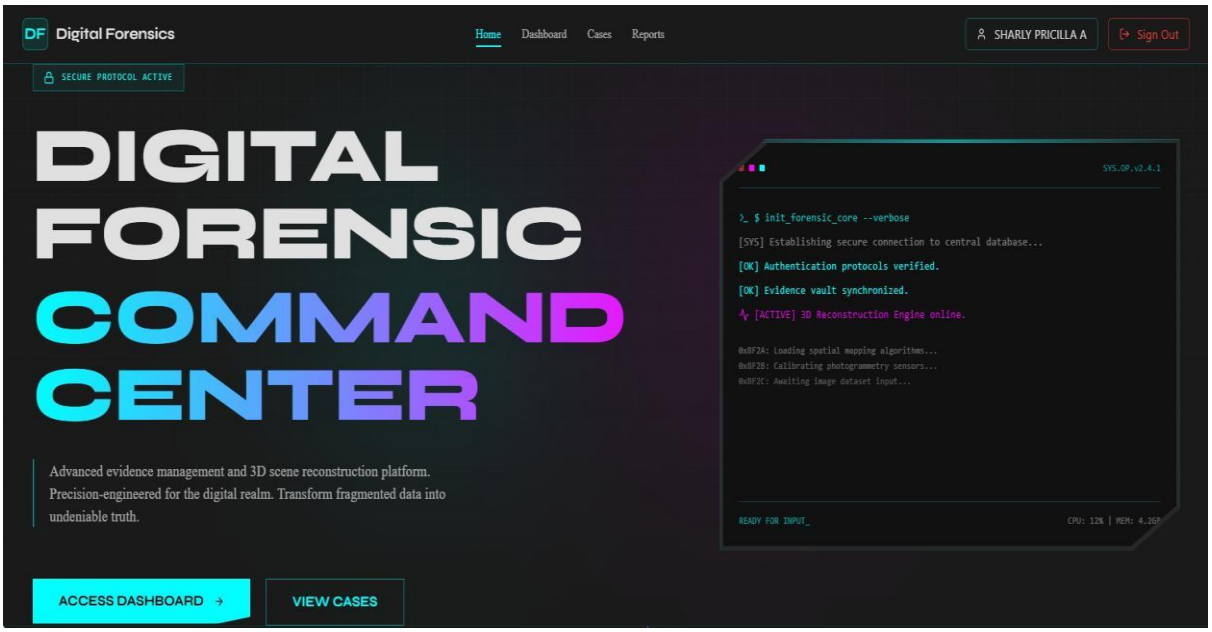

Figure B.3 Home Page

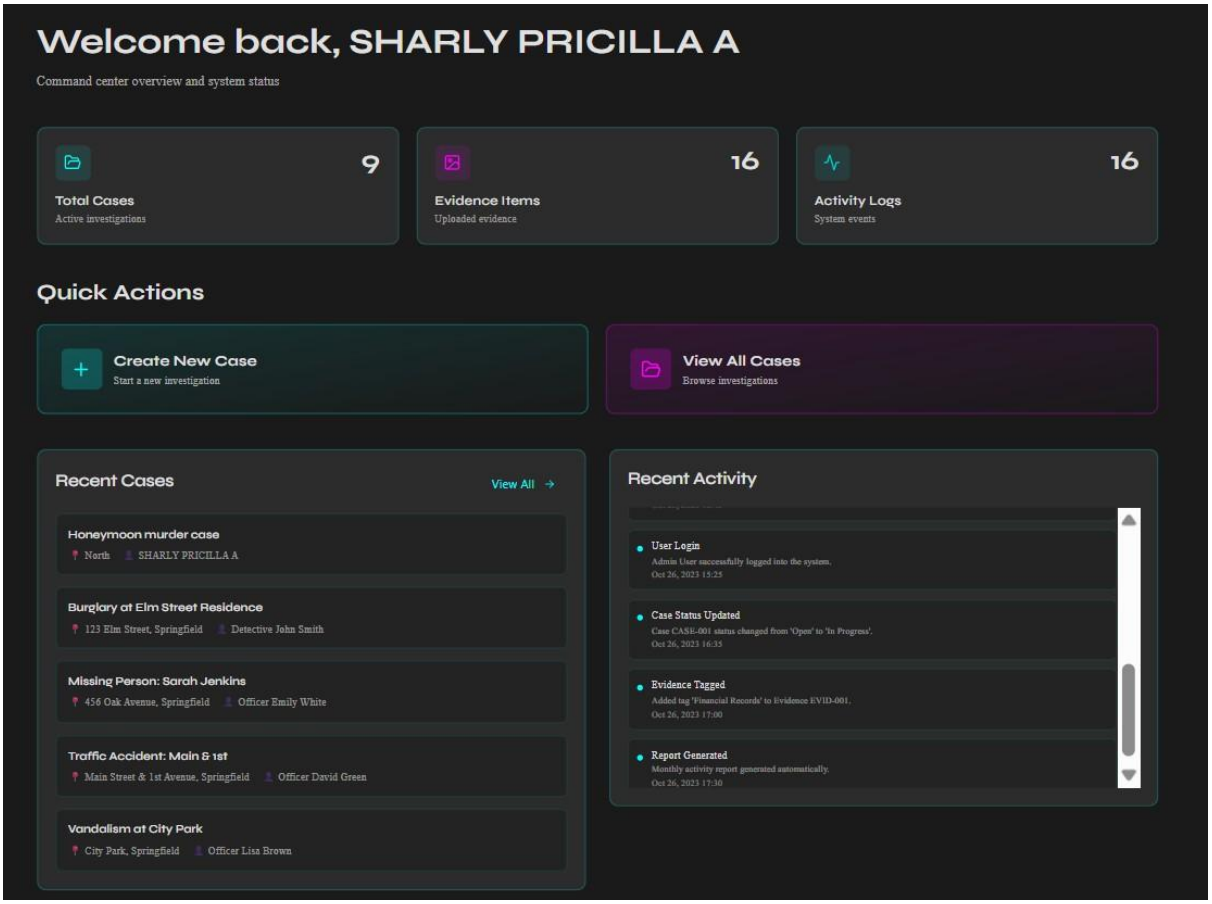

Figure B.4 Dashboard Page

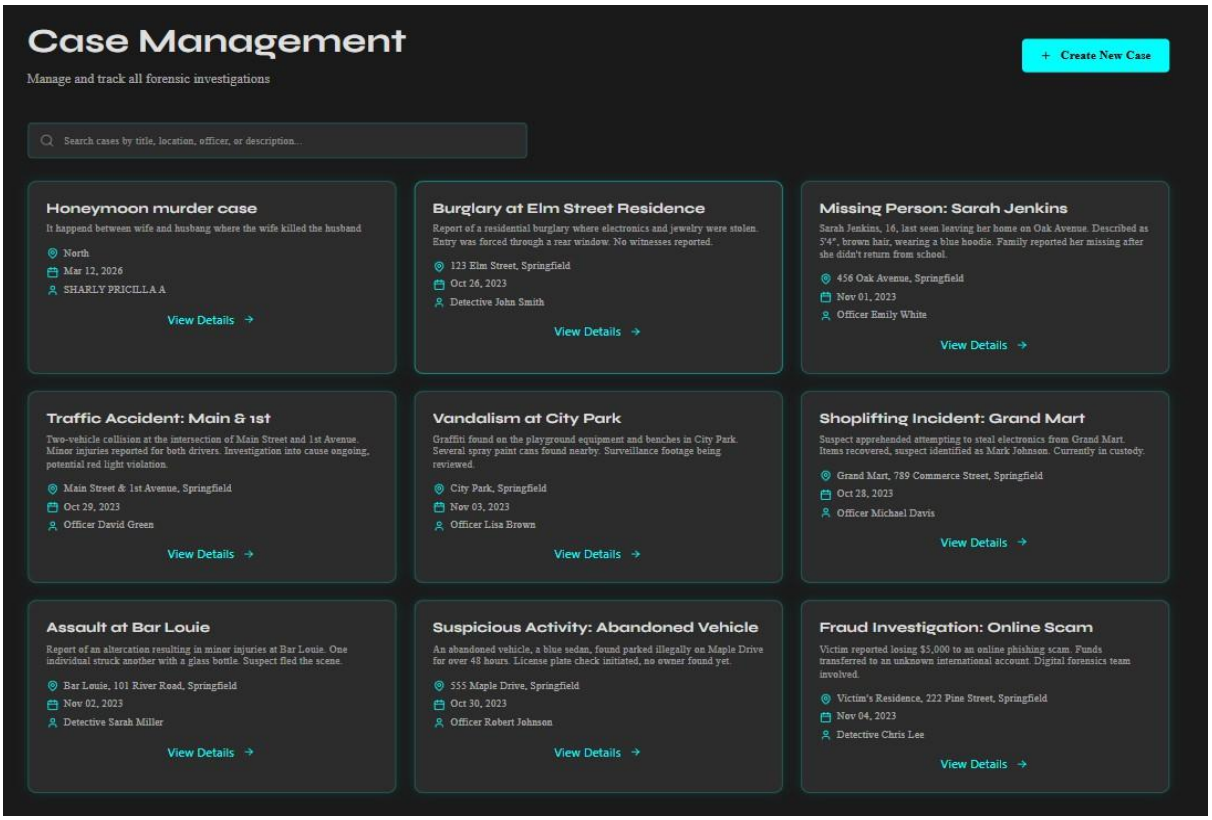

Figure B.5 Case Management Page

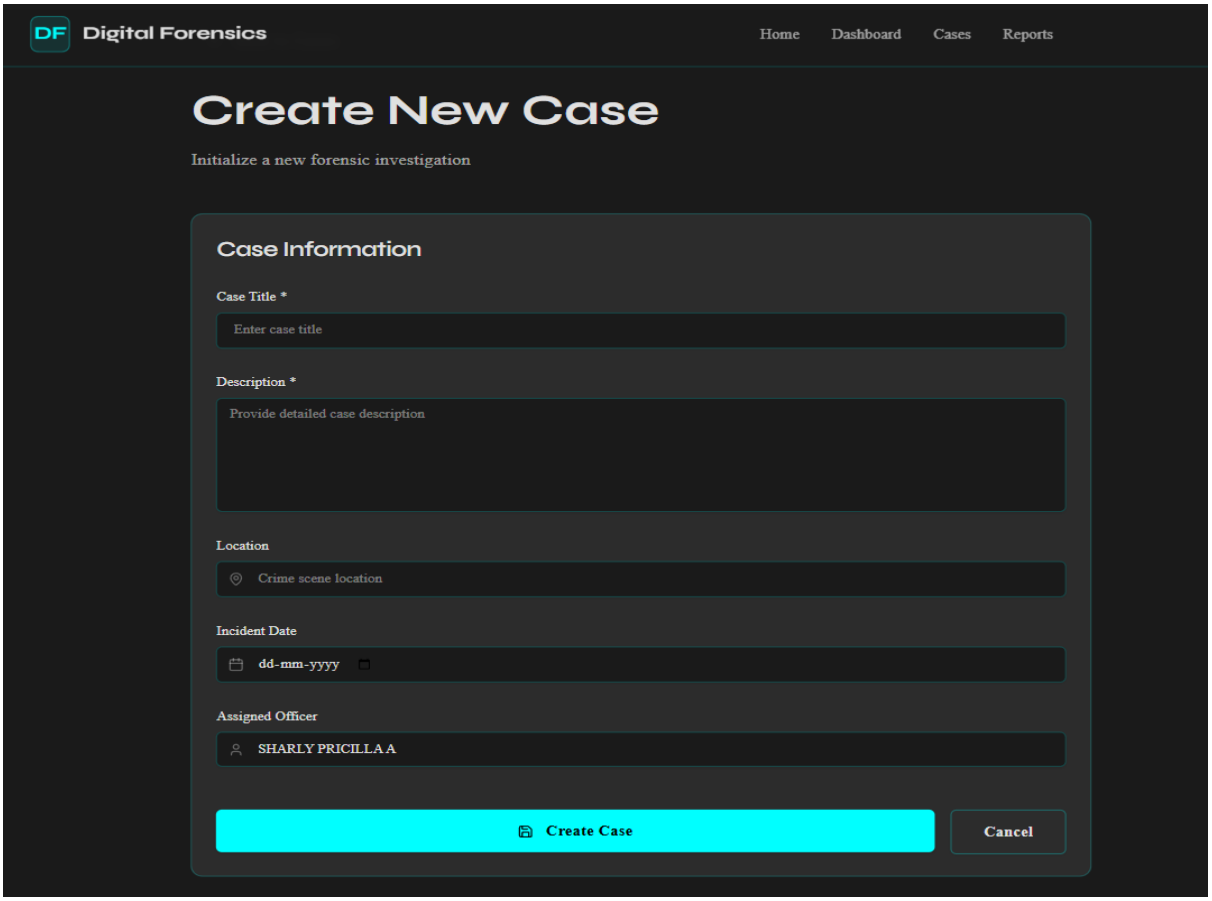

Figure B.6 New Case Page

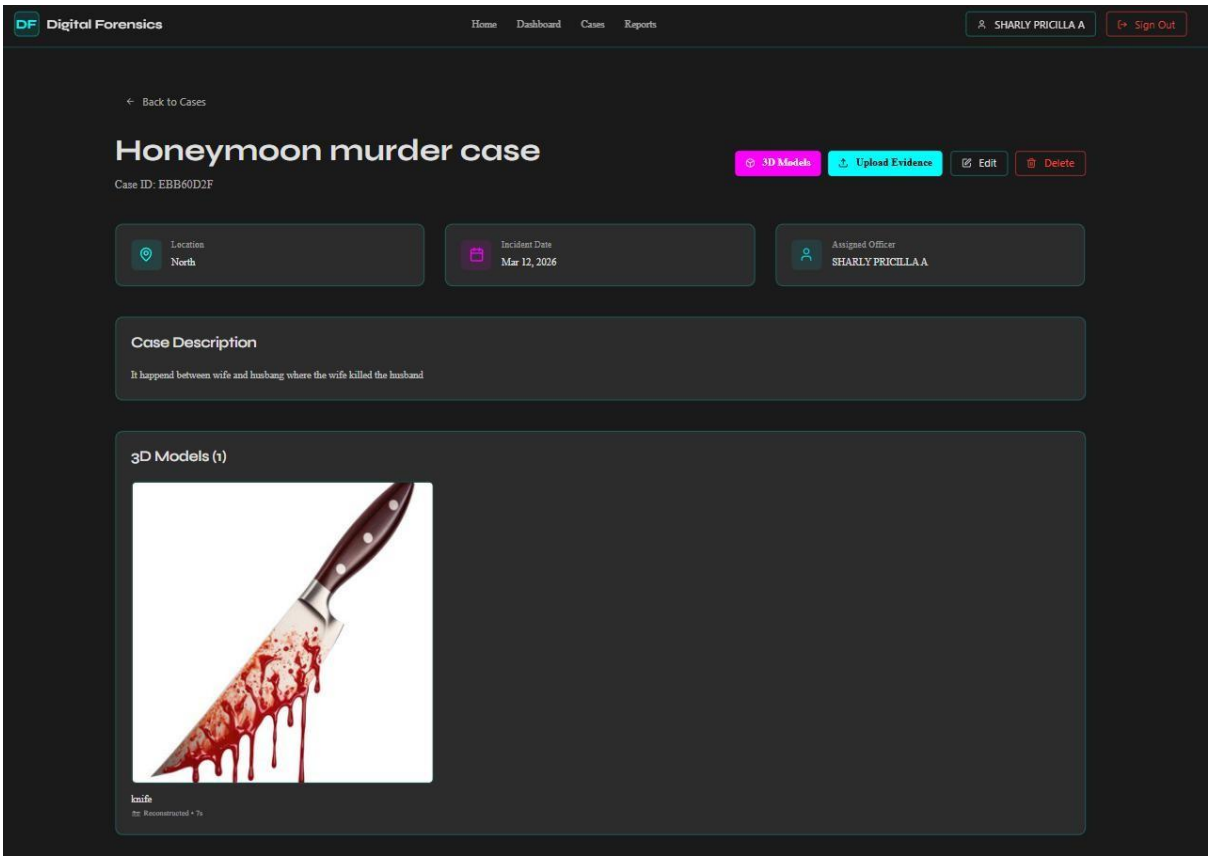

Figure B.7 Case Details Page

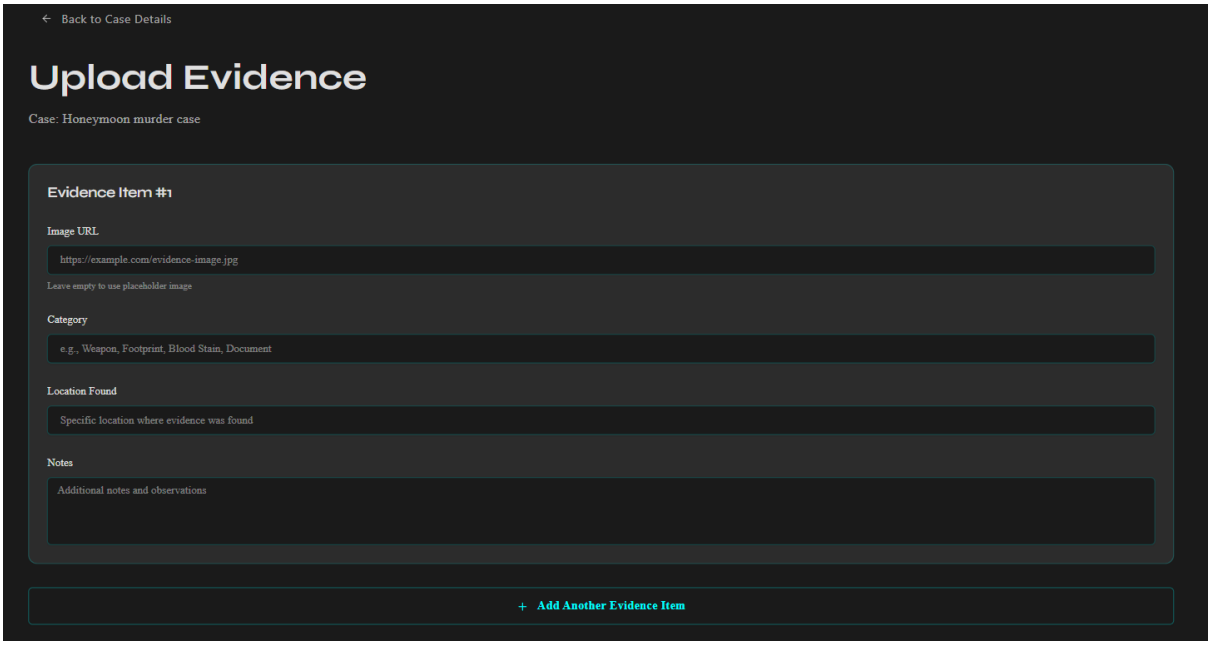

Figure B.8 Evidence Upload Page

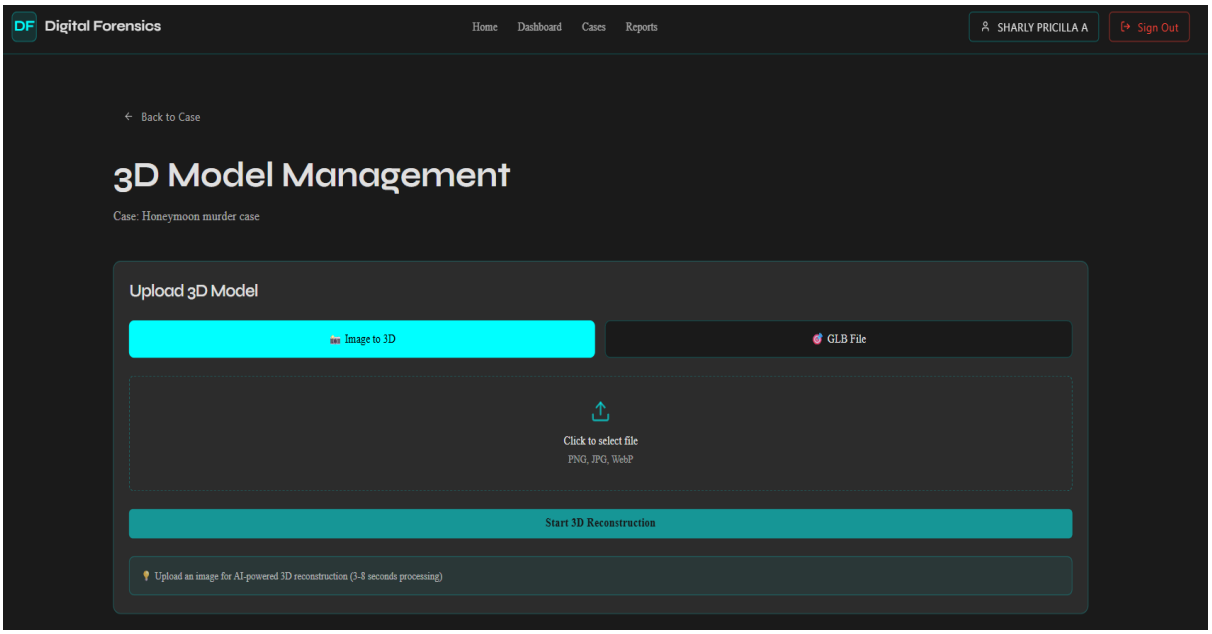

Figure B.9 3D Model Management Page

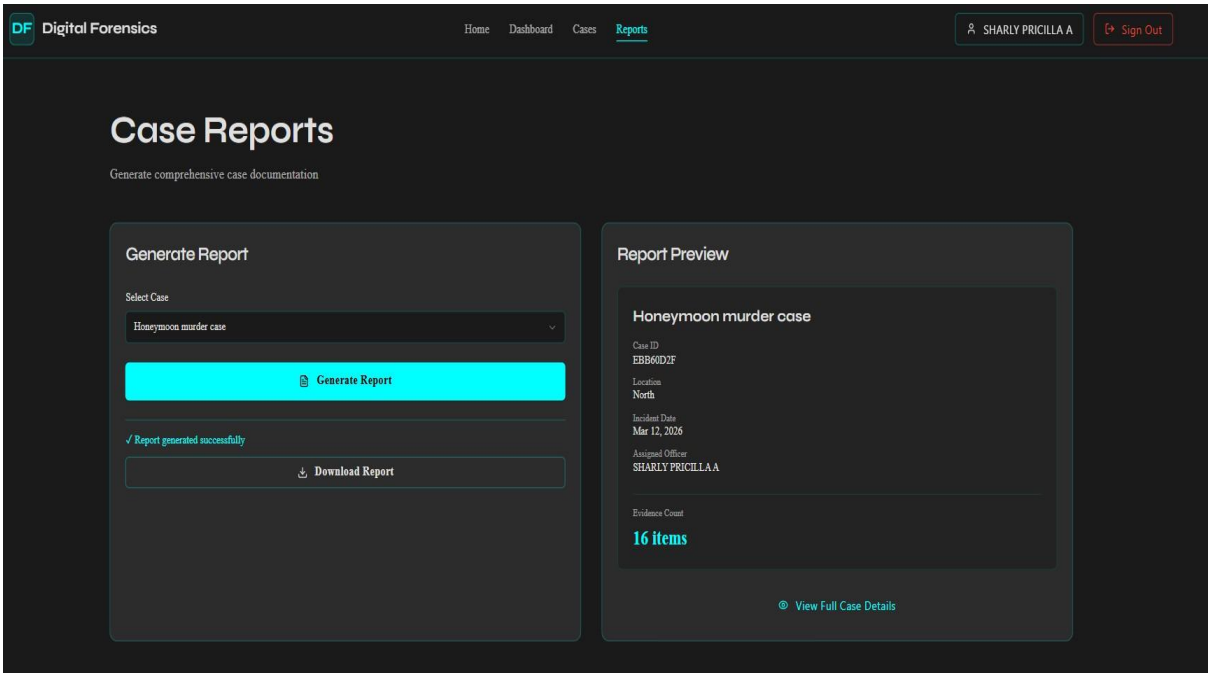

Figure B.10 Case Report Page

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